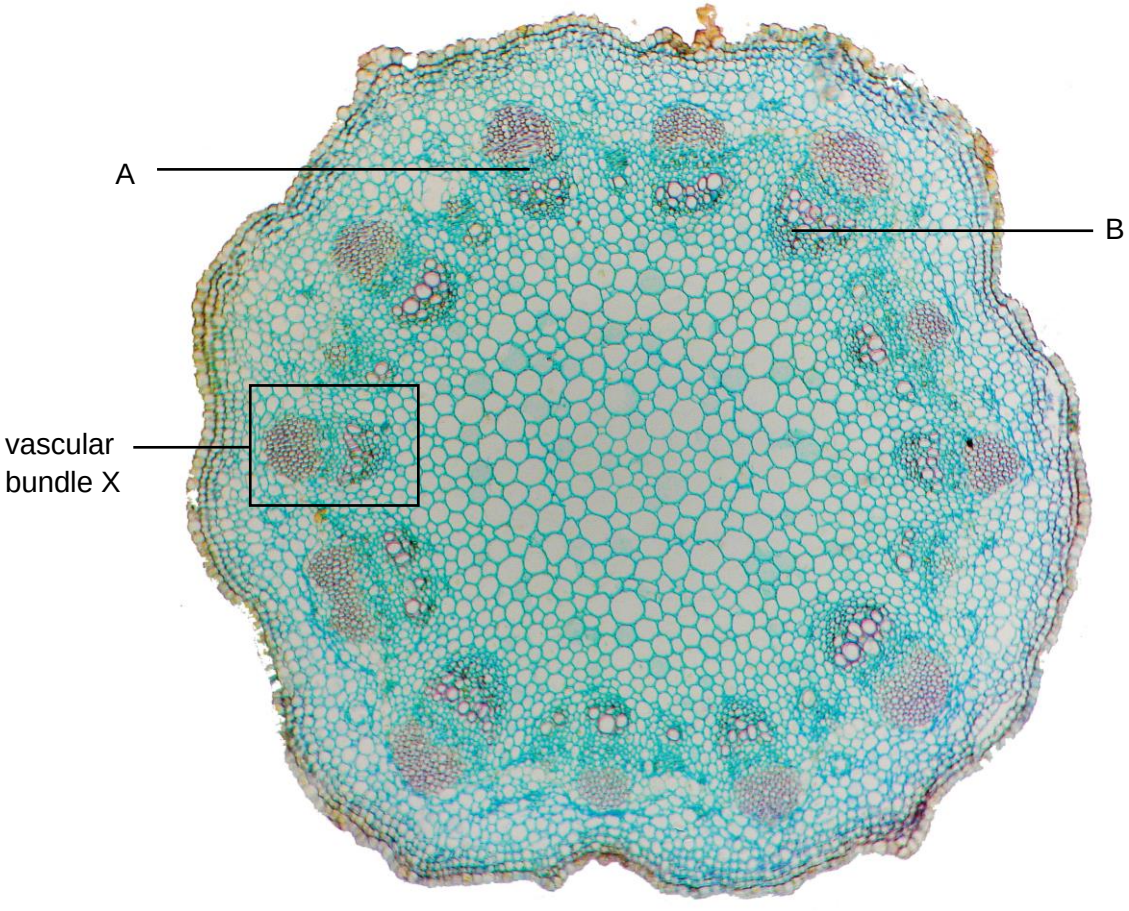
Structured questions

|!|BCELA0110000001|!| ★★

John observed an unknown plant tissue using a microscope. The photomicrograph below shows the plant tissue observed by John.



Plant tissue, T.S. (×30)

(a) John suggested that it is a stem tissue but not a root tissue. Based on the photomicrograph, give ***two*** observable features which support John's claim.

(2 marks)

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(b) Describe how structure A and structure B are structurally adapted to their functions in transport. (4 marks)

(c) John would like to study vascular bundle X in detail. Describe the steps he should take to observe vascular bundle X using high power magnification. (3 marks)

##

(a) The vascular bundles are arranged in a circle near the periphery of the tissue. (1)
The xylem is located in the inner region while the phloem is located in the outer region. (1)

(b) Sieve tubes present in structure A contain cytoplasm but no nuclei, (1)
which allows flow of nutrients with little resistance. (1)
Structure B has no end wall / is hollow, (1)
which allows flow of water with little resistance. (1)

(c) Adjust the position of the slide so that vascular bundle X is at the centre of the field of view. (1)
Rotate the nosepiece to select an objective of higher magnification. (1)
Adjust the fine adjustment knob until vascular bundle X is in focus. (1)

(9 marks)

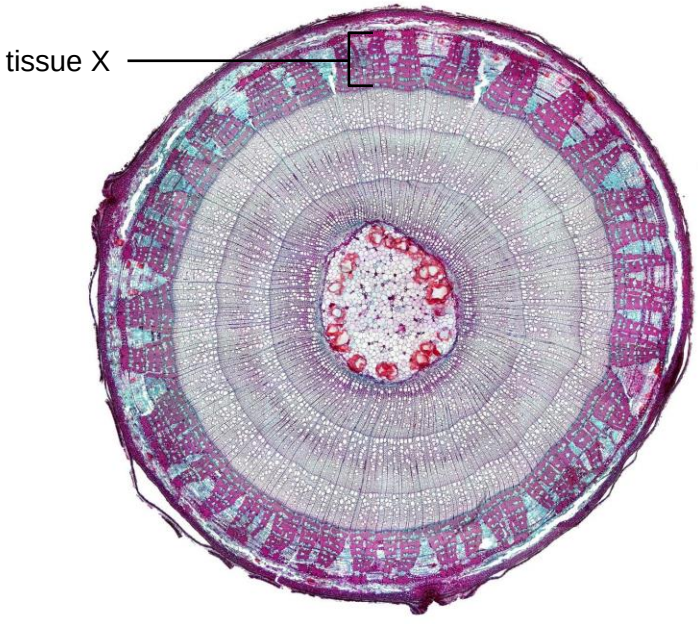
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	4
(c)	Practical skills	3

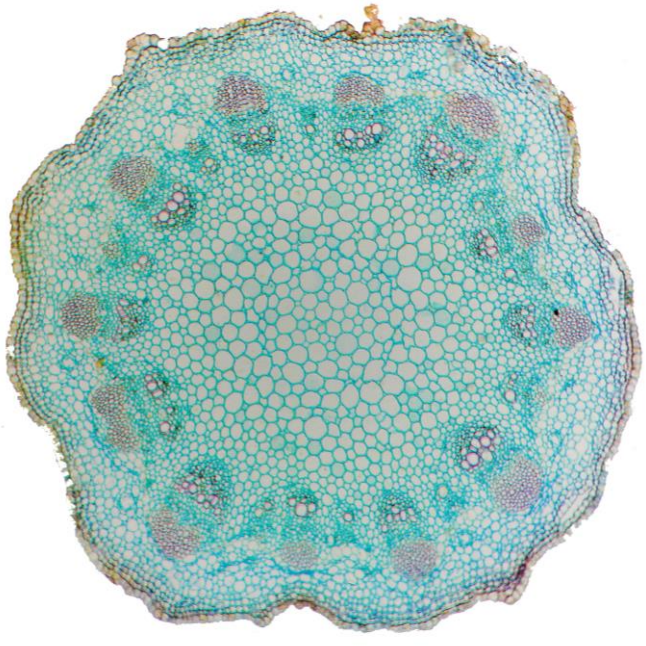
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|!|BCELB0110000002|!| ★

The photomicrographs below show the transverse sections of the stems of plants A and B.



Stem of plant A, T.S. (×10)



Stem of plant B, T.S. (×15)

(a) Label tissue X in the photomicrograph of plant B. (1 mark)

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(b) (i) Some insects insert their tube-like mouthparts into tissue X in the plant stem to feed. Give ***one*** organic substance present in the fluid in tissue X. (1 mark)

(ii) State how and where the organic substance in (i) is formed. (2 marks)

(iii) Where is this organic substance transported to? Give ***two*** functions of this organic substance in plants. (3 marks)

(c) On a hot and sunny day, plant A and plant B were exposed under sunlight for five hours. Predict the appearance of plant A and plant B, and explain your answer with respect to the support mechanism in the stem of each plant.

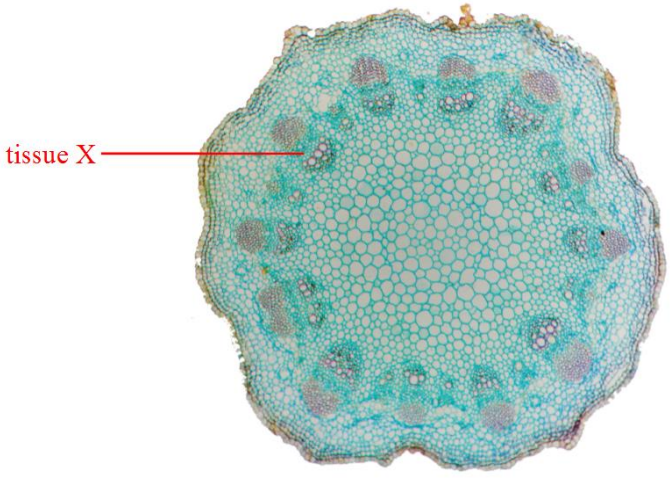
(i) Plant A (3 marks)

(ii) Plant B (4 marks)

##

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(a) Correct position (1)



(b) (i) Sucrose (1)

(ii) The organic substance in (i) is produced through the process of photosynthesis (1)

in the leaves. (1)

(iii) It is transported to the roots / fruits / storage organs. (1)

It can serve as a respiratory substrate to release energy. (1)

It can be used to form structural materials (cellulose) for formation of new cells. (1)

(c) (i) Plant A remains upright. (1)

The support of the stem of plant A mainly relies on the rigidity of xylem (or secondary xylem). (1)

The xylem of plant A is lignified / contains thick-walled cells to provide mechanical support to the plant. (1)

(ii) Plant B wilts. / The stem of plant B bends. (1)

The support of the stem of plant B mainly relies on the turgidity of thin-walled cells (cortex and pith cells). (1)

When plant B is exposed under sunlight, the transpiration rate becomes higher than the rate of water absorption. (1)

There is net water loss and the cells lose turgidity and become flaccid, the plant therefore loses support and bends. (1)

(14 marks)

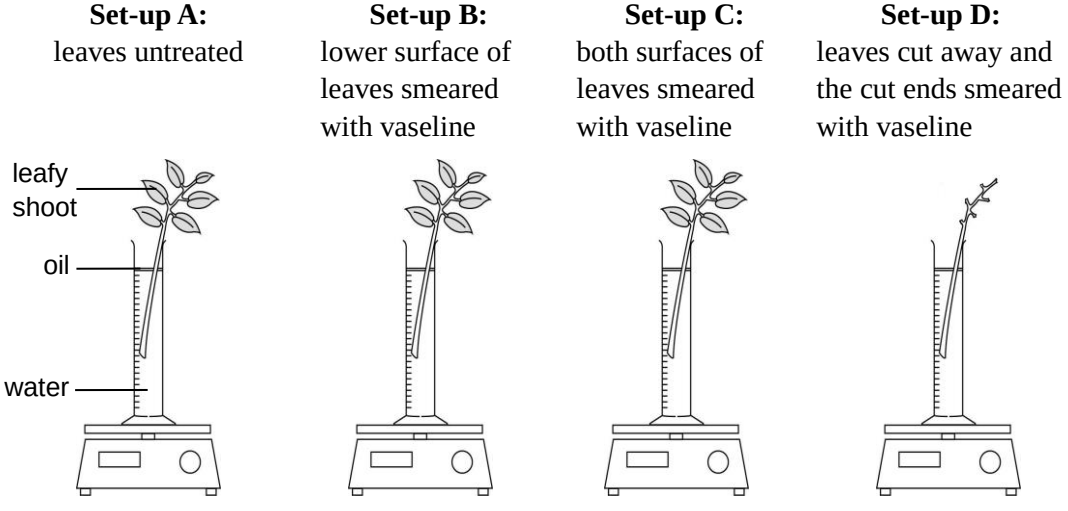
##

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	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	3
(c)(i)	Applying concepts	3
(c)(ii)	Applying concepts	4

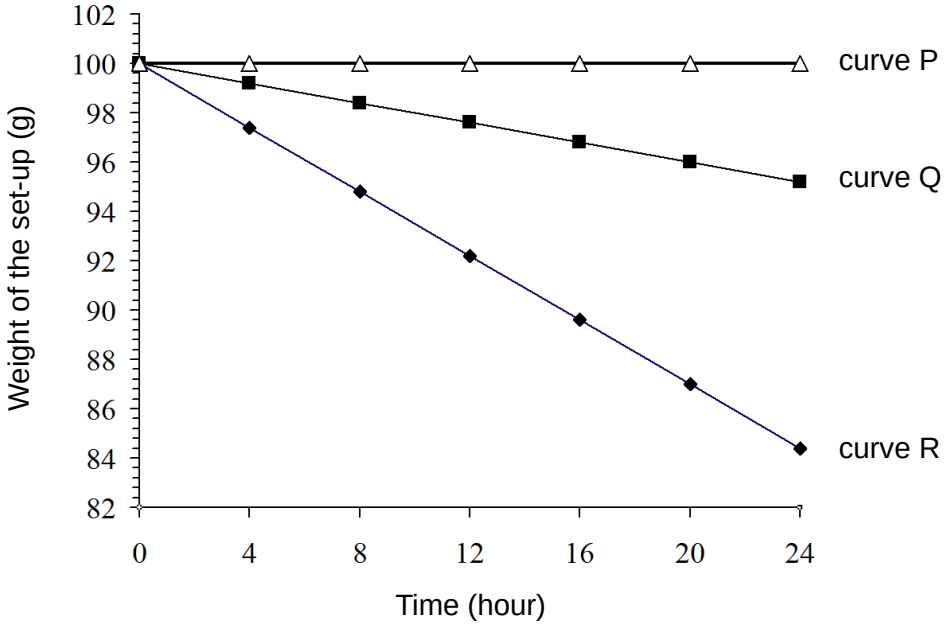
|!|BCELB0110000003|!| ★ **SI**

In an investigation to study transpiration, four similar leafy shoots with the same number of leaves were treated in different ways as shown below:



The weight of each set-up was adjusted to 100 g at the beginning of the investigation. The weights of the set-ups were then recorded at regular intervals for 24 hours. The graph below shows the changes in weight during the investigation.

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(a) Curve P represents the weight changes of two of the set-ups. State which the two set-ups are. Explain your answer. (4 marks)

(b) State and explain which curve represents the weight changes in each of the other set-ups. (4 marks)

(c) Based on the results, deduce the relative abundance of stomata on the upper and lower epidermis of the leaves. (2 marks)

##

(a) Set-ups C and D (1)
There is no change in their weights. No transpiration occurred (1)
because in set-up C the stomata are covered by vaseline (1)
while in set-up D no leaves are present for transpiration. (1)
(b) Curve R represents the weight changes in set-up A. (1)
Curve Q represents the weight changes in set-up B. (1)
In set-up B, half of the leaf surface is covered by vaseline. (1)
Therefore, the transpiration rate of set-up B is slower than that of set-up A. (1)
(c) More stomata are present on the lower epidermis than the upper epidermis. (1)
When the lower epidermis is covered by vaseline, the rate of transpiration is greatly reduced / is reduced by more than half. (1)

(10 marks)

##

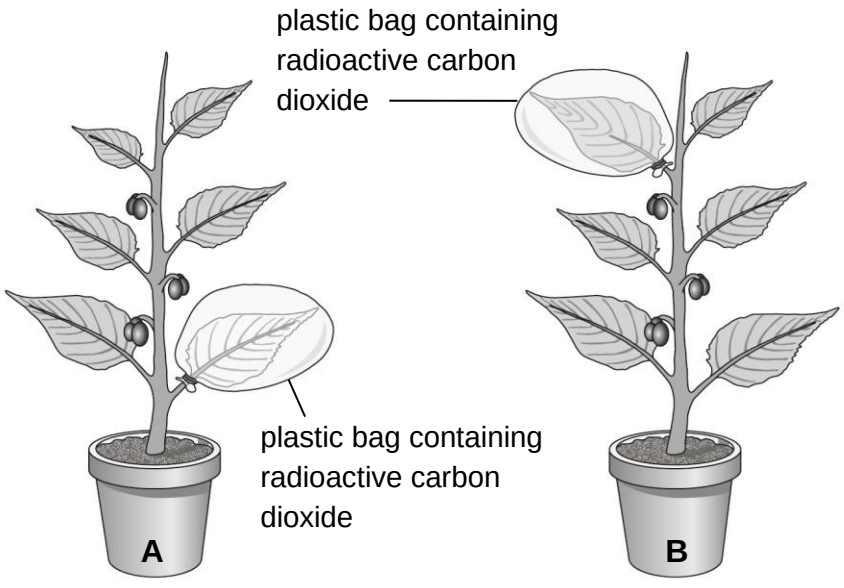
	Skills assessed	Marks
(a)	Analysing data	4
(b)	Analysing data	4
(c)	Analysing data	2

|!|BCELA0110000004|!| ★★ SI

A student carried out an experiment to study the translocation of carbohydrate in two tomato plants as follows:

1. On each plant, the student chose a leaf and wrapped it in a plastic bag containing radioactive carbon dioxide.
2. The two plants were left under sunlight for two hours.
3. The student removed the plastic bags and then left the plants undisturbed for 24 hours.
4. The radioactivity of different parts of the plants was measured.

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(a) Write a simple word equation to represent the overall process of photosynthesis. (1 mark)

_____

(b) Suggest why the plants were left undisturbed for 24 hours before measuring their radioactivity. (1 mark)

_____

_____

(c) The amount of radioactivity in various parts of the plants is recorded in the table below.

	Radioactivity (arbitrary unit)	
Parts	**Plant A**	**Plant B**
Untreated leaf	150	210
Radioactive CO_2 treated leaf	11500	11600
Stem	1200	1020
Fruit	4650	8800
Roots	2600	820

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(i) Compare and explain the difference in radioactivity between the stem and fruit. (3 marks)

(ii) What conclusion can be drawn based on the results of the two plants? (2 marks)

##

(a) carbon dioxide + water $\xrightarrow[\text{chlorophyll}]{\text{light energy}}$ carbohydrates + oxygen (1)

(b) To allow sufficient time for the radioactive photosynthetic products to be transported to other parts of the plants. (1)

(c) (i) The radioactivity of the fruit is higher than that of the stem (1)
because fruits store carbohydrates (1)
while the stem consumes carbohydrates in respiration as it grows. (1)

(ii) Radioactivity can be detected in the parts above and below the radioactive CO_2 treated leaves. (1)
This shows that the transport of carbohydrates is bidirectional. (1)

(7 marks)

##

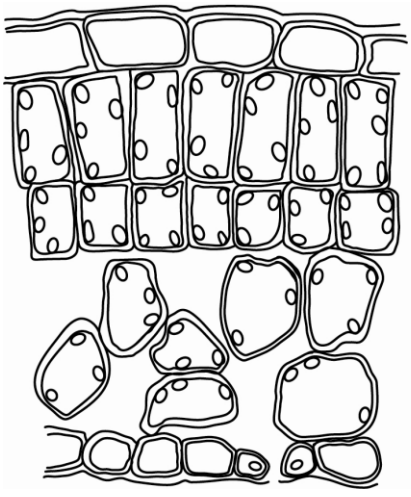
	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	1
(c)(i)	Analysing data	1
	Applying concepts	2
(c)(ii)	Applying concepts	2

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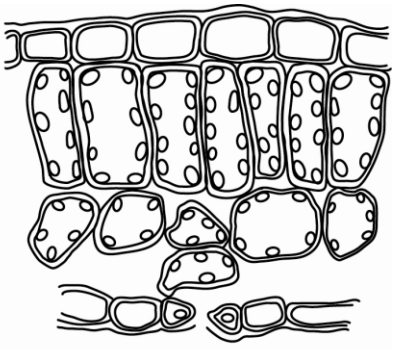
|!|BCELB0110000005|!| ★

The following diagrams show the transverse sections of leaf X and leaf Y. Leaf X is developed in sunlight and leaf Y is developed in shade. The two diagrams are of the same magnification.

Leaf X



Leaf Y



(a) With reference to the diagrams, describe ***three*** structural differences between leaf X and leaf Y. (3 marks)

(b) Describe and explain how leaf X and leaf Y are adapted to their growing environments. (4 marks)

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(c) With reference to the diagrams, state ***two*** ways by which the leaves are adapted for reducing water loss. (2 marks)

_____

_____

##

(a) Any ***three*** of the following comparisons: (1 mark each) (3)

Leaf X	Leaf Y
- two layers of cells in palisade mesophyll	- one layer of cells in palisade mesophyll
- more cells	- less cells
- fewer chloroplasts per cell	- more chloroplasts per cell
- thicker	- thinner
- more air spaces	- less air spaces

(b) Leaf X has more air spaces and this increases the rate of gas exchange / carbon dioxide uptake (1)

as carbon dioxide may limit the rate of photosynthesis under bright light condition. (1)

Leaf Y has more chloroplasts per cells and so the plant can get a sufficient amount of light energy for photosynthesis (1)

as the light available is limited under shady condition. (1)

(Or other acceptable answers)

(c) Cuticle is present on the leaf surfaces. (1)

Stomata are only present on the lower epidermis / Stomata are absent on the upper epidermis. (1)

_____

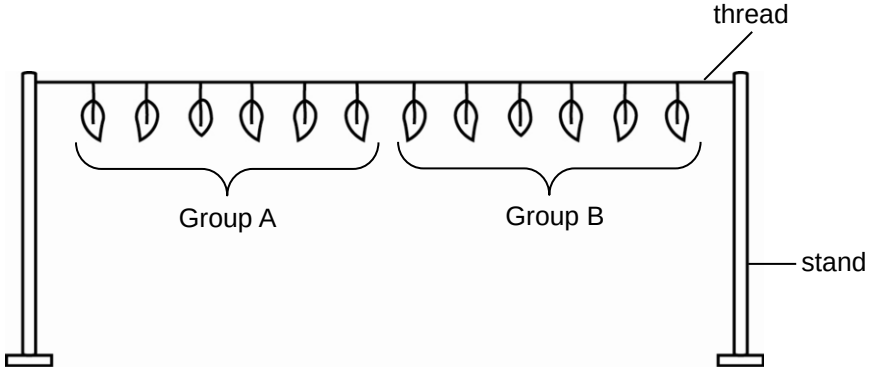
(9 marks)

##

	Skill assessed	Marks
(a)	Analysing data	3
(b)	Applying concepts	4
(c)	Analysing data	2

|!|BCELB0110000006|!| ★ **SI**

Alice carried out an experiment to investigate the rate of water loss from leaves. She obtained 12 similar leaves from a dicotyledonous plant. She left six of them untreated (group A) and six of them with the upper surface smeared with vaseline (group B). She weighed the leaves and tied all the leaves on a thread as shown below.

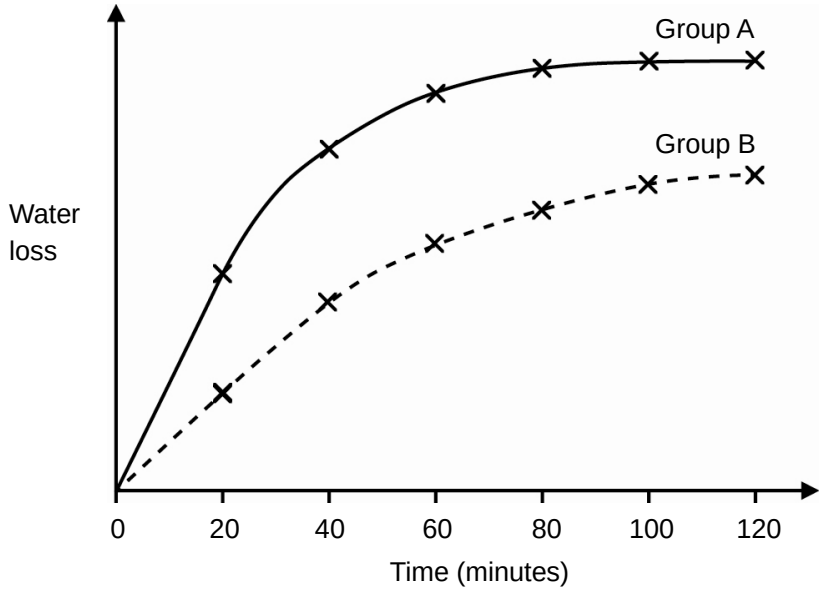


(a) State ***two*** environmental factors that Alice needed to be kept constant during the experiment. (2 marks)

_____

_____

(b) During the experiment, Alice weighed the leaf from each group every 20 minutes over a period of two hours. The decrease in leaf mass can represent the amount of water loss. The following graph shows the results.



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(i) According to the graph shown above, explain the change in the mass of leaves of group A. (3 marks)

(ii) Explain for the differences between the results of the leaves in group A and group B. (2 marks)

(c) If a leaf from group A was put in a beaker of hot water, bubbles were released from the leaf surfaces.

(i) What would be the difference in the number of bubbles released from both surfaces? (1 mark)

(ii) Explain why bubbles were released. (2 marks)

##

(a) Any ***two*** of the following: (2)
- Light intensity (1)
- Relative humidity (1)
- Wind speed (1)
- Temperature (1)

(b) (i) Due to the water potential gradient between the inside of the leaf and the atmosphere, (1)
water evaporates out through the stomata. (1)
As water potential gradient decreases during the experiment, stomata close to prevent excessive water loss. (1)

(ii) Since the stomata on the upper surface are blocked by vaseline, (1)
less water is evaporated / transpired form the leaves. (1)

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(c) (i) The lower surface would release more bubbles. (1)

(ii) The air in the intercellular spaces expanded if the leaf was put in hot water. (1)

The expanded air escaped through the stomata as bubbles. (1)

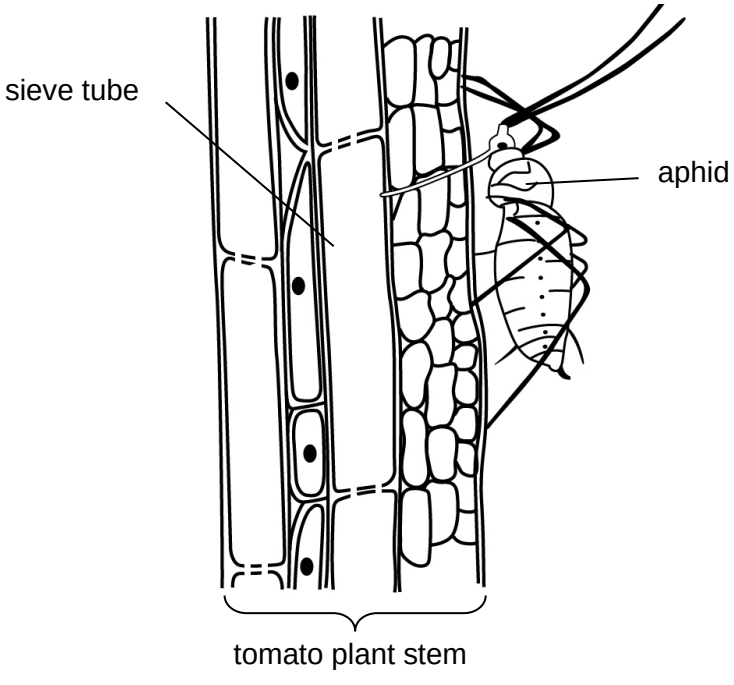
(10 marks)

##

	Skill assessed	Mark(s)
(a)	Practical skills	2
(b)(i)	Analysing data	1
	Applying concepts	2
(b)(ii)	Analysing data	1
	Applying concepts	1
(c)(i)	Applying concepts	1
(c)(ii)	Applying concepts	2

|!|BCELB0110000007|!| ★ SI

The diagram below shows an aphid feeding on plant juice. (The aphid and the plant stem are not drawn to the same scale.)



In an investigation, an aphid feeding on plant juice is anaesthetized with a high concentration of carbon dioxide. The body of the aphid is cut off, leaving only its mouthparts in the stem. Liquid flows out of the cut end of the mouthparts.

(a) What substances, besides water, could be found in the liquid flowing out from the mouthparts? (2 marks)

(b) Name the process by which the substances in (a) are transported in plants. Where are these substances transported to? (3 marks)

(c) What do the results of this investigation show? (2 marks)

##

(a) Sugars / carbohydrates / sucrose (1)
and amino acids (1)
(b) Translocation (1)
To the growing points in the shoots / roots. (1)
To storage organs / fruits. (1)
(c) The organic nutrients in plants are transported (1)
in the phloem tissues. (1)

(7 marks)

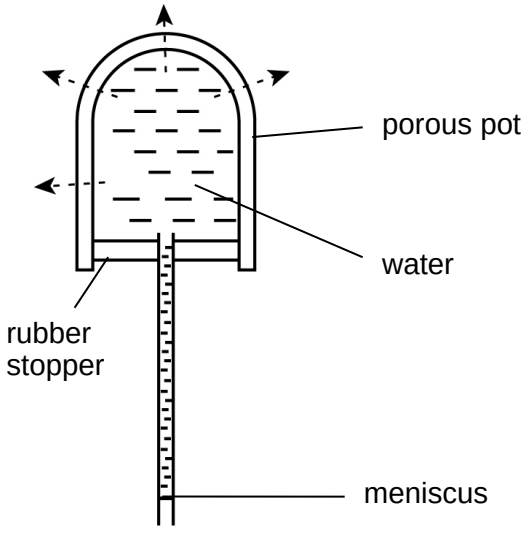
##

	Skill assessed	Mark(s)
(a)	Applying concepts	2
(b)	Understanding basic concepts	3
(c)	Applying concepts	2

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|!|BCELB0110000008|!| ★

The diagram below shows an apparatus set up to demonstrate a process that occurs in plants.



(a) Name the process demonstrated by the apparatus. (1 mark)

_____

(b) State the structures in a plant that are represented by:

(i) the pores in the pot (1 mark)

_____

(ii) the porous pot (1 mark)

_____

(iii) the glass tube (1 mark)

_____

(c) Suggest ***two*** environmental conditions that could speed up the rate of the movement of the meniscus up the glass tube. (2 marks)

_____

_____

_____

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(d) What would happen to the movement of the meniscus if the porous pot is covered by a plastic bag? Explain your answer. (4 marks)

##

(a) Transpiration (1)

(b) (i) Stomata (1)

(ii) Leaf (1)

(iii)Xylem vessel (1)

(c) Increase in temperature (1)

Decrease in humidity of surrounding air (1)

(d) The movement of the meniscus slows down. (1)

because water vapour is trapped inside the plastic bag. (1)

The air humidity inside the bag increases. (1)

The rate of evaporation of water decreases as the concentration gradient of water vapour decreases. (1)

(10 marks)

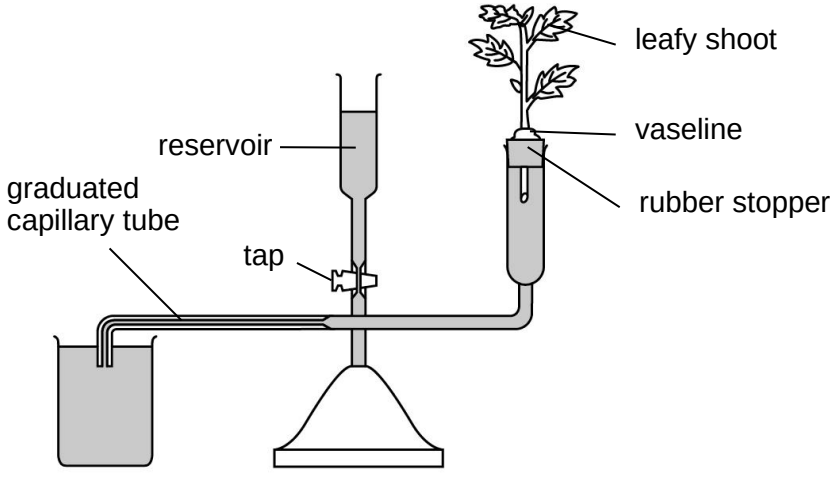
##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(b)(iii)	Analysing data	1
(c)	Applying concepts	2
(d)	Applying concepts	4

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|!|BCELB0110000009|!| ★ **SI**

The diagram below shows a bubble potometer.



(a) Describe how you would make use of this apparatus to investigate the effect of wind speed on the transpiration rate of the leafy shoot. (4 marks)

(b) Give ***one*** assumption for using this potometer to measure the transpiration rate of the leafy shoot. (1 mark)

(c) What precautions should be taken during this experiment? (3 marks)

##

(a) Introduce an air bubble into the graduated capillary tube by lifting the open end of the tube out of water and then returning it to water. (1)

Carry out the experiment under still air first. (1)

Measure the distance moved by the air bubble over a set period of time. (1)

Carry out a series of experiments at different wind speeds using a fan set at different speeds or distances from the potometer. (1)

(b) The rate of transpiration is the same as the rate of water absorption. (1)

(c) Any ***three*** of the following: (3)

- The leafy shoot should be cut in water to prevent air from entering and blocking the xylem vessels of the stem. (1)
- Allow the shoot to equilibrate to new conditions before taking any readings. (1)
- Use the same leafy shoot for the whole experiment. (1)
- All conditions other than wind speed (e.g. temperature, light intensity, humidity, etc.) should be kept the same throughout the experiment. (1)

(8 marks)

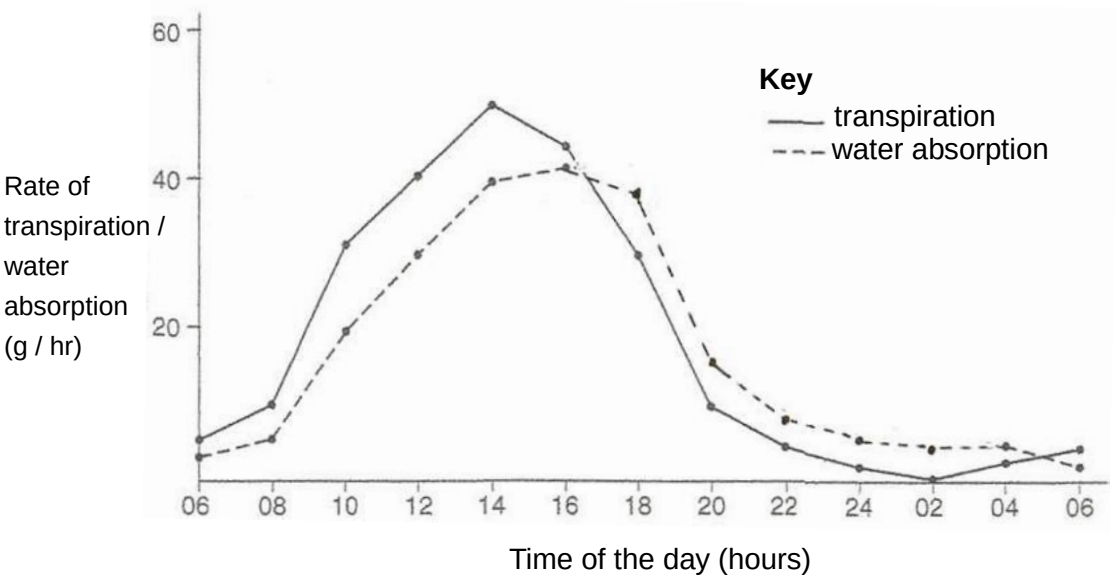
##

	Skill assessed	Mark(s)
(a)	Practical skills	4
(b)	Practical skills	1
(c)	Practical skills	3

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|!|BCELB0110000010|!| ★

The graph below shows the results of an investigation of the rate of transpiration and absorption of water from a herbaceous plant on a clear and hot summer day.



(a) Describe and explain the changes in the rate of transpiration from 0800 to 1400. (4 marks)

(b) Compare the changes in the rate of transpiration to that of the rate of water absorption. (2 marks)

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(c) If the patterns of transpiration and water absorption shown in the graph continued for a few days, what would happen to the plant? (3 marks)

_____

_____

##

(a) From 0800 to 1400, the rate of transpiration increases rapidly (1)
because the light intensity increases during daytime and the stomata open more widely. (1)
Sunlight also raises the air temperature. (1)
The evaporation of water from the leaf surface is faster. (1)

(b) Any ***two*** of the following: (2)
- Both curves have a similar shape. (1)
- Absorption peak occurs four hours after transpiration peaks. (1)
- Peak of transpiration is slightly higher. (1)

(c) As water absorption cannot keep up with transpiration, the plant would have a net loss in water. (1)
The plant would lose its support from turgidity. (1)
The plant would eventually wilt. (1)

(9 marks)

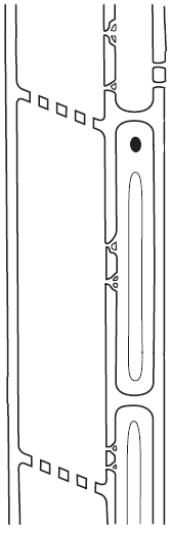
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
	Applying concepts	3
(b)	Analysing data	2
(c)	Analysing data	1
	Applying concepts	2

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|!|BCELB0110000011|!| ★

The diagram below shows the longitudinal section of the phloem tissue of a plant.



(a) Label sieve tube, sieve plate and companion cell in the diagram. (3 marks)

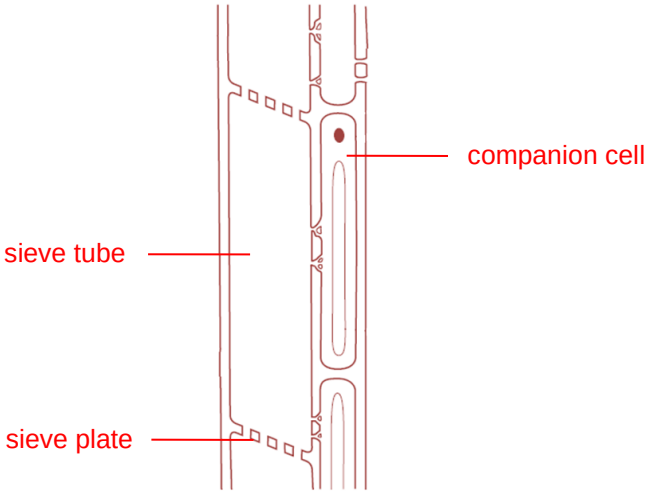
(b) What is the function of the sieve tube? (1 mark)

(c) Give ***one*** similarity and ***two*** differences between the structure of a xylem vessel and a sieve tube. (5 marks)

(d) Why is the companion cell important to the sieve tube? (2 marks)

##

(a) Correct labelling (3)



(b) To transport organic nutrients / sucrose and amino acids within the plant. (1)

(c) Similarity: They have no nuclei. (1)

Differences:

Xylem vessels do not have cytoplasm, (1)

while sieve tubes have cytoplasm. (1)

Xylem vessels do not have end walls, (1)

while sieve tubes have end walls called sieve plates. (1)

(d) Companion cells have cytoplasm, a nucleus and numerous mitochondria. (1)

They provide energy to support the metabolic activities of the sieve tube. (1)

(11 marks)

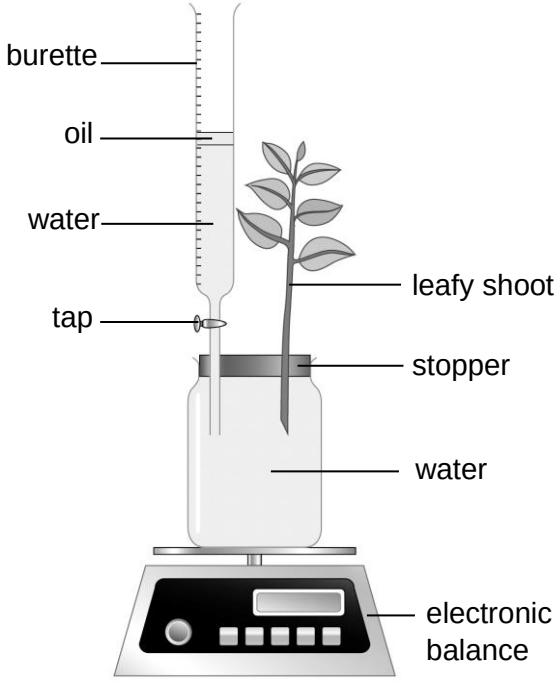
##

	Skill assessed	Mark(s)
(a)	Analysing data	3
(b)	Understanding basic concepts	1
(c)	Applying concepts	5
(d)	Understanding basic concepts	2

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|!|BCELB0110000012|!| ★ **SI**

The diagram below shows the set-up used to measure the transpiration rate of a leafy shoot.



(a) Explain why the lower end of the leafy shoot should be cut under water before the experiment. (1 mark)

_____

(b) What is the function of the oil? (1 mark)

_____

(c) The table below shows the reading of the electronic balance and the burette after the set-up was left under bright light for five hours.

	Reading of the electronic balance (g)	**Reading of the burette (cm^3)**
Initial reading	200.0	40.0
Final reading	197.5	37.0

(Given that the density of water is 1 g / cm^3)

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(i) Calculate the amount of water absorbed by the shoot. (2 marks)

(ii) Calculate the amount of water lost by the shoot. (1 mark)

(iii) Describe and explain the significance in the difference between the amount of water absorbed and lost by the shoot. (3 marks)

(d) Explain how the transpiration rate of the shoot would change if the set-up is placed in a dimly lit room. (2 marks)

##

(a) To prevent the blockage of xylem by air bubble. (1)

(b) To prevent the evaporation of water. (1)

(c) (i) Amount of water absorbed = the change in the water level in the burette

$= (40.0 – 37.0)\ cm^3$ (1)

$= 3.0\ cm^3$ (1)

(ii) Amount of water lost = the change in the weight of the whole set-up

$= (200.0 – 197.5)\ g \div 1\ g\ /cm^3$

$= 2.5\ cm^3$ (1)

(iii) The amount of water absorbed is larger than the amount of water lost by the leafy shoot. (1)

There is a net gain in water by the shoot. (1)

Water is retained in the plant for use in photosynthesis, growth and other metabolic activities. (1)

(d) The transpiration rate decreases. (1)

It is because in darkness, the stomata close. (1)

(10 marks)

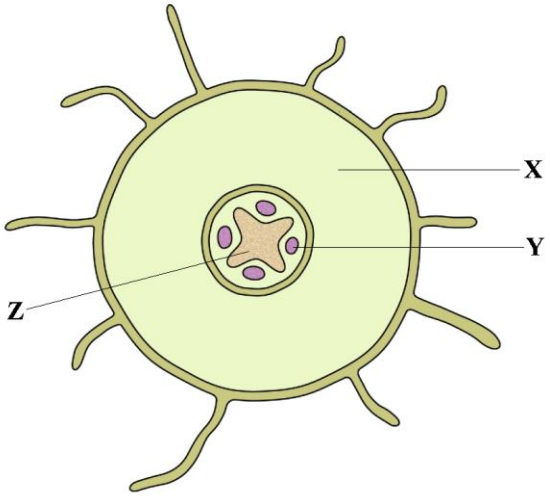
##

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	Skill assessed	Mark(s)
(a)	Practical skills	1
(b)	Practical skills	1
(c)(i)	Mathematical skills	2
(c)(ii)	Mathematical skills	1
(c)(iii)	Applying concepts	3
(d)	Applying concepts	2

|!|BCELC0110000013|!|

The diagram below shows the cross section of a plant organ.



(a) Name this plant organ as shown above. (1 mark)

(b) Identify structures X and Y. (2 marks)

(c) State and explain ***two*** adaptive features of this plant organ to absorb water and minerals. (4 marks)

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(d) State ***two*** functions of structure Z. (2 marks)

_____

_____

##

(a) Root (1)

(b) Structure X: cortex (1)

Structure Y: phloem (1)

(c) The plant root is not covered by cuticle, (1)

hence water and mineral ions can enter easily. (1)

Many epidermal cells develop root hairs (1)

which greatly increase the surface area for water and minerals uptake / which are long and fine to penetrate the spaces between soil particles for absorption. (1)

(Or other acceptable answers)

(d) Structure Z (xylem) is responsible for upward transport of water and minerals. (1)

Structure Z (xylem) provides mechanical support to the plant. (1)

(9 marks)

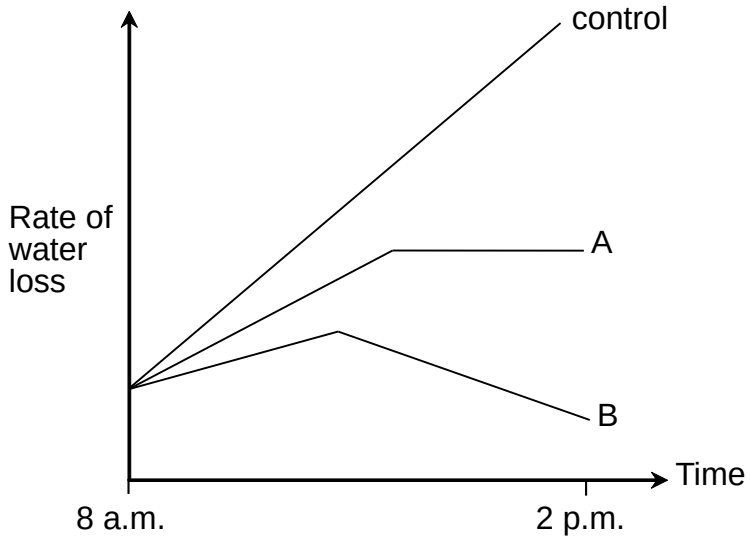
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Analysing data	2
(c)	Analysing data	2
	Applying concepts	2
(d)	Understanding basic concepts	2

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|!|BCELB0110000014|!| ★

The graph below shows the rates of water loss in plants under different conditions over a period of time. Curves A and B show the results of plants that were either enclosed in an airtight plastic bag or exposed to an outdoor environment. In the control, a plant was exposed in normal conditions, with sufficient light illumination.



(a) (i) What is the scientific term used to describe 'water loss' in plants? (1 mark)

_____

(ii) State ***two*** significance of such process to plants. (2 marks)

_____

_____

(b) Which curve shows the rate of water loss of a plant that was enclosed in an airtight plastic bag? Explain your answer. (5 marks)

_____

_____

_____

_____

_____

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(c) Describe the other curve and suggest ***one*** condition which causes differences in the results when compared with the control. (3 marks)

##

(a) (i) Transpiration (1)

(ii) Any ***two*** of the following: (2)

- It produces a cooling effect to prevent the plant from being overheated. (1)
- It helps the transport of water and minerals upwards. (1)
- It helps absorb water and minerals from the soil. (1)

(Or other acceptable answers)

(b) Curve B (1)

As the water vapour lost by transpiration is trapped inside the plastic bag, (1)

the relative humidity inside the bag increases, (1)

this decreases the water potential gradient between the air space in leaves and the surrounding air and hence the rate of diffusion of water vapour. (1)

Thus the rate of water loss through diffusion decreases after a while as shown by curve B. (1)

(c) The rate of water loss first increases (1)

and then remains constant after reaching the highest point. (1)

Condition: a cloudy day (when the light intensity does not increase any further)

(Or other acceptable answers) (1)

(11 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)	Analysing data	1
	Applying concepts	4
(c)	Analysing data	2
	Applying concepts	1

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|!|BCELB0110000015|!| ★

The following figures show the structures of two xerophytes. ***Fig. 1*** shows a cactus and ***Fig. 2*** shows the transverse section of the leaf of plant X, which has very long roots.

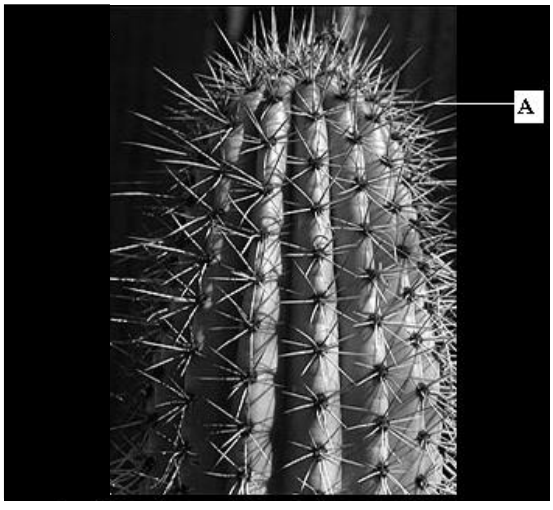


Fig. 1

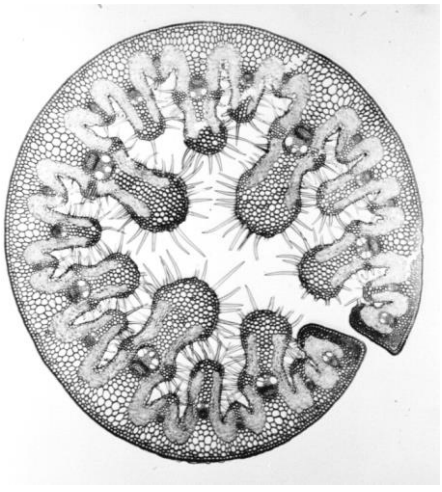


Fig. 2

(a) With reference to the cactus,

(i) to which organ does structure A belong? (1 mark)

_____

(ii) explain the significance of the shape of structure A to the cactus. (2 marks)

_____

_____

(b) With reference to plant X,

(i) in which habitat does plant X grow? (1 mark)

_____

(ii) explain how the leaves and roots of plant X are adapted to the habitat answered in (b)(i). (3 marks)

_____

_____

_____

##

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(a) (i) Leaf (1)

(ii) The needle-shaped leaf can reduce the leaf surface area (1)
so that the transpiration rate is lower to reduce water loss. (1)

(b) (i) in dry habitats / deserts (1)

(ii) The leaf is rolled with the stomata on the inside of the leaf (1)
This helps trap moisture inside the leaf and thus reduces water loss by transpiration. (1)
Long roots can absorb water from deep soil. (1)

(7 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
	Applying concepts	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
	Applying concepts	2

|!|BCELB0110000016|!| ★

Kelly stopped watering her potted plant and the plant wilted as shown below. She then watered the plant. After two hours, the plant became upright.



(a) Suggest a reason to explain the wilting appearance. (1 mark)

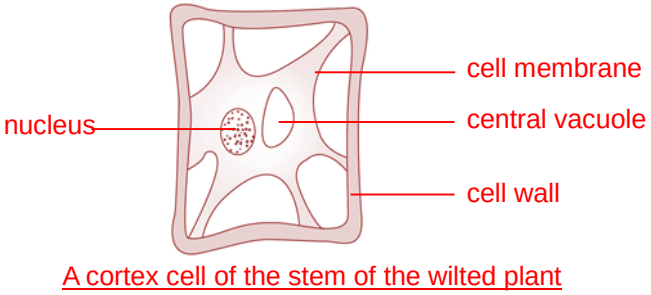
(b) Draw a labelled diagram to show a cortex cell found in the stem of the wilted plant. (3 marks)

(c) Explain how watering can make the plant become upright. (3 marks)

##

(a) The rate of transpiration is higher than the rate of water absorption. (1)

(b)



A cortex cell of the stem of the wilted plant

Correct labels (any three) (1)
Correct drawing of a flaccid cell (1)
Correct title (1)

(c) When water supply is adequate, the thin-walled cells in the cortex gain water by osmosis. (1)

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These cells become turgid and press against each other. (1)

The turgidity of these cells supports the stem to stand upright. (1)

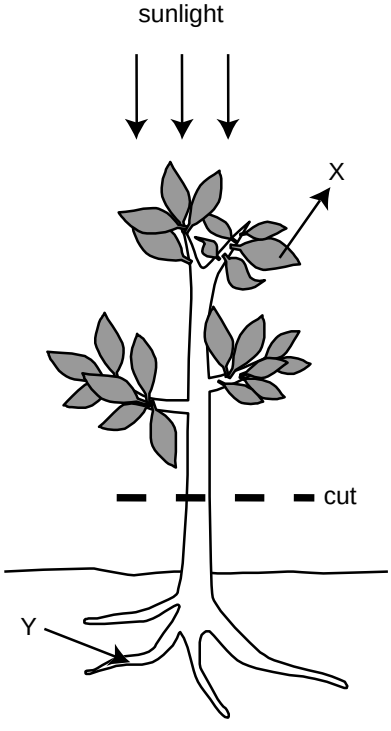
(7 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Presenting information	3
(c)	Applying concepts	3

|!|BCELC0110000017|!|

The diagram below shows a young plant exposed to sunlight. Arrows X and Y represent the directions of water movement.



(a) Describe the water movement as represented by arrow X. (4 marks)

(b) A cut is made on the stem of the plant as shown in the diagram in order to make a transverse section for observation under microscope. Draw a labelled diagram of the transverse section. Shade the part in which water travels along. (4 marks)

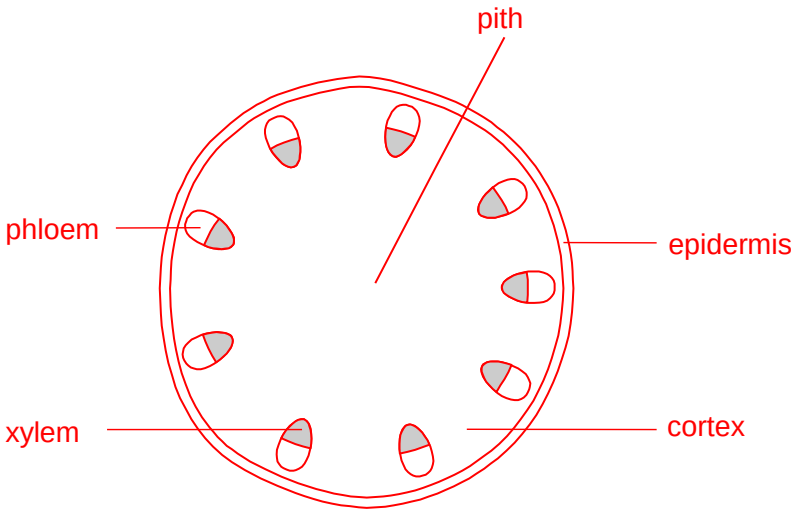
(c) Describe how water is transported from arrow Y to arrow X through the part you shaded in (b). (3 marks)

##

(a) Transpiration takes place at arrow X. (1)
Water evaporates from the surface of mesophyll cells into the air spaces (1)
and diffuses from the air spaces into the atmosphere through the stomata (1)
when the water vapour concentration in the air spaces is higher than that in the atmosphere. (1)

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(b)



Transverse section of the stem

Correct drawing (1)

Correct labels (any four) and title (2)

Correct shading of the xylem (1)

(c) Water lost to the air spaces at arrow X is continuously replaced by water in the mesophyll cells. (1)

Water is drawn from the neighbouring cells and the xylem vessels by osmosis. (1)

Due to the surface tension of water, a force is produced which pulls water up the xylem vessels from the root (arrow Y) to the leaf. (1)

(11 marks)

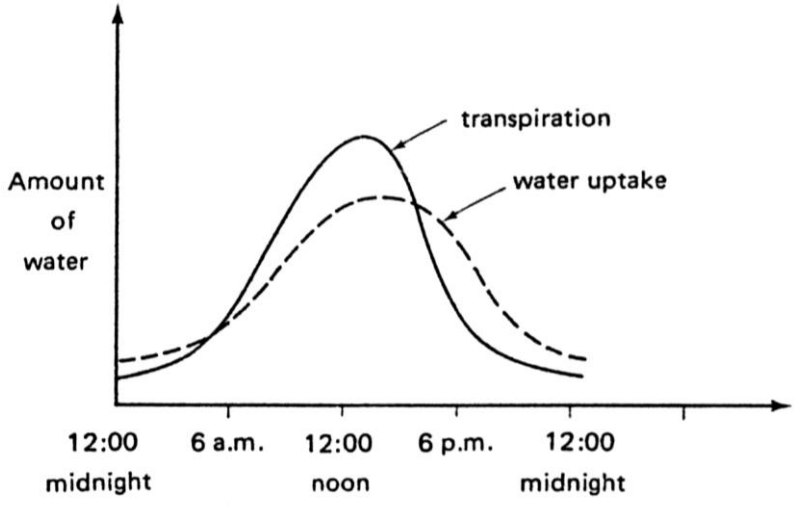
##

	Skill assessed	Marks
(a)	Understanding basic concepts	4
(b)	Presenting information	4
(c)	Understanding basic concepts	3

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|!|BCELB0110000018|!| ★

The graph below shows the amount of water lost by transpiration and taken up by a potted plant on a sunny day.



(a) Explain why the amount of water taken up changes with that lost by transpiration and slightly lags behind. (1 mark)

(b) How would the results change if the measurements were taken on a rainy day? (1 mark)

(c) Use the graph to find out the time(s) at which

(i) the highest rate of transpiration occurred. (1 mark)

(ii) the rate of transpiration was equal to that of water uptake. (1 mark)

(d) According to the graph, explain why the water content of the plant was getting higher from 4 p.m. to 12:00 midnight. (2 marks)

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(e) On the whole, the total amount of water taken up by the plant is greater than that of water lost. Suggest ***three*** ways in which the plant uses the water it retains.

(3 marks)

##

(a) This is because the plant has to set up a transpiration pull for the upward transport of water, which enhances the absorption of water at the roots. (1)

(b) Both the amounts of water lost and taken up by the plant would become lower on a rainy day. (1)

(c) (i) 1:00 p.m./ 1:30 p.m. (1)

(ii) 5:00 a.m. and 3:30 p.m. / 4:00 p.m. (1)

(d) From 4 p.m. to 12:00 midnight, the amount of water taken up was greater than that lost by transpiration. (1)

This is because stomata were closed during the night and no transpiration occurred through the stomata. (1)

(e) Water is incorporated into the cytoplasm and is used to make new cells. (1)

Water can be used as a reactant in photosynthesis. (1)

Water acts as a medium for transport of minerals and carbohydrates. (1)

(Or other acceptable answers)

(9 marks)

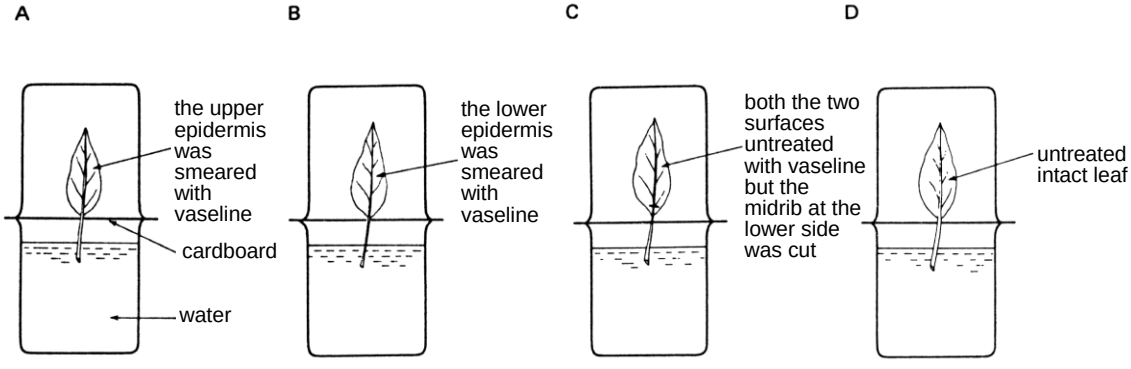
##

	Skill assessed	Mark(s)
(a)	Applying concepts	1
(b)	Applying concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Analysing data	1
(d)	Analysing data	1
	Applying concepts	1
(e)	Understanding basic concepts	3

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|!|BCELB0110000019|!| ★ **SI**

The following diagram shows an experiment. In the experiment, four similar leaves were treated differently. The leaves were fixed in pieces of perforated cardboard and placed over four water-filled beakers. Another beaker was covered on top of each leaf.



(a) Which two beakers would first become misty on the inner walls? Explain your answers. (4 marks)

(b) How can you show that the mist contains water? (1 mark)

(c) Explain why the other two beakers had the inner walls remained clear. (4 marks)

(d) State ***two*** precautions of this experiment. (2 marks)

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##

(a) beaker A and beaker D (1)

This is because the lower epidermis of their leaves is not smeared with vaseline. (1)

Most of the stomata are present on the lower epidermis, (1)

and water is lost through the stomata during transpiration. (1)

(b) To test if the liquid on the inner wall of the beaker will turn the dry cobalt chloride paper pink. (1)

(c) In beaker B, the stomata on the lower epidermis of the leaf had been blocked by vaseline. (1)

Transpiration occurred slowly. (1)

In beaker C, the xylem of the leaf had been cut and (1)

water could not be transported to the leaf and lost through transpiration. (1)

(d) The leaf stalk should be cut under water. (1)

The cut end of leaf stalk should be immersed completely in the water. (1)

(or other reasonable answers)

(11 marks)

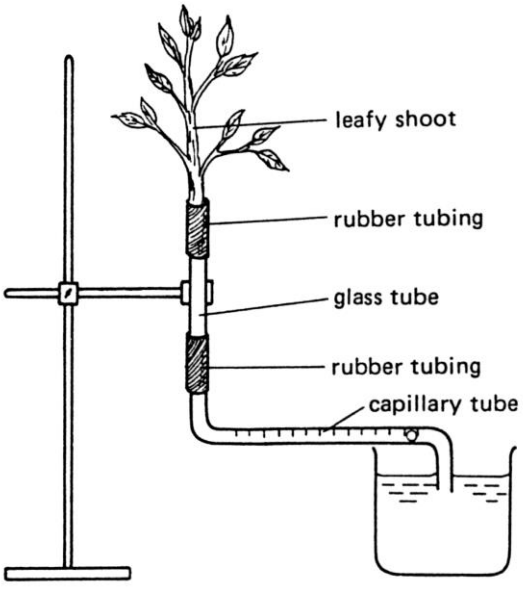
##

	Skill assessed	Mark(s)
(a)	Analysing data	1
	Applying concepts	3
(b)	Practical skills	1
(c)	Applying concepts	4
(d)	Practical skills	2

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|!|BCELB0110000020|!| ★ **SI**

The following diagram shows an apparatus used in measuring the rate of water loss from a leafy shoot. The apparatus was placed in a normal condition.



(a) Name this apparatus. (1 mark)

(b) State an assumption that has to be made in this experiment. (1 mark)

(c) State and explain a precaution which should be taken in cutting the leafy shoot. (2 marks)

(d) Suggest ***two*** limitations in using this apparatus to find out the water loss from the leafy shoot. (2 marks)

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(e) Describe and explain the effect of the following changes on the distance travelled by the air bubble in the capillary tube.

(i) a fan is placed near the apparatus (3 marks)

(ii) all the lighting is switched off (2 marks)

##

(a) Bubble potometer (1)

(b) The rate of water uptake is equal to the rate of transpiration. (1)

(c) The leafy shoot should be cut under water (1)
to prevent the entry of air bubbles which may block the xylem vessels. (1)

(d) Any ***two*** of the following: (2)

- The potometer is too small to fit a larger plant / the whole plant. (1)
- The potometer does not have a sufficient volume to fit the extensive root system of a plant. (Without the root system the rate of water absorption measured is lower than the normal rate.) (1)
- There is friction between the air bubble and the capillary wall. (1)

(e) (i) The distance travelled by the air bubble increases as wind speed increases. (1)
This is because wind removes water vapour from leaf surface and (1)
this maintains a steep concentration gradient for the diffusion of water vapour from the air spaces in the leaf to the surrounding air. (1)

(ii) The distance travelled by the air bubble decreases as light intensity decreases. (1)
This is because stomata close in darkness and so no water in the air spaces can diffuse out. (1)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Practical skills	1
(c)	Practical skills	2
(d)	Practical skills	2
(e)(i)	Applying concepts	3
(e)(ii)	Applying concepts	2

|!|BCELA0110000021|!| ★★ **SI**

CCEA GCE AS Biology Assessment Unit AS 2 Jun 2014 Q7

A bubble potometer was used to investigate the rate of water uptake by a leafy shoot from a young sycamore tree.

A hairdryer was used to investigate the effect of wind strength and environmental temperature on the rate of movement of the trapped bubble. The temperature was varied by selecting either the hot or cold setting on the hairdryer, and the wind strength was varied by changing the distance between the hairdryer and the shoot.

The results are shown in the table below.

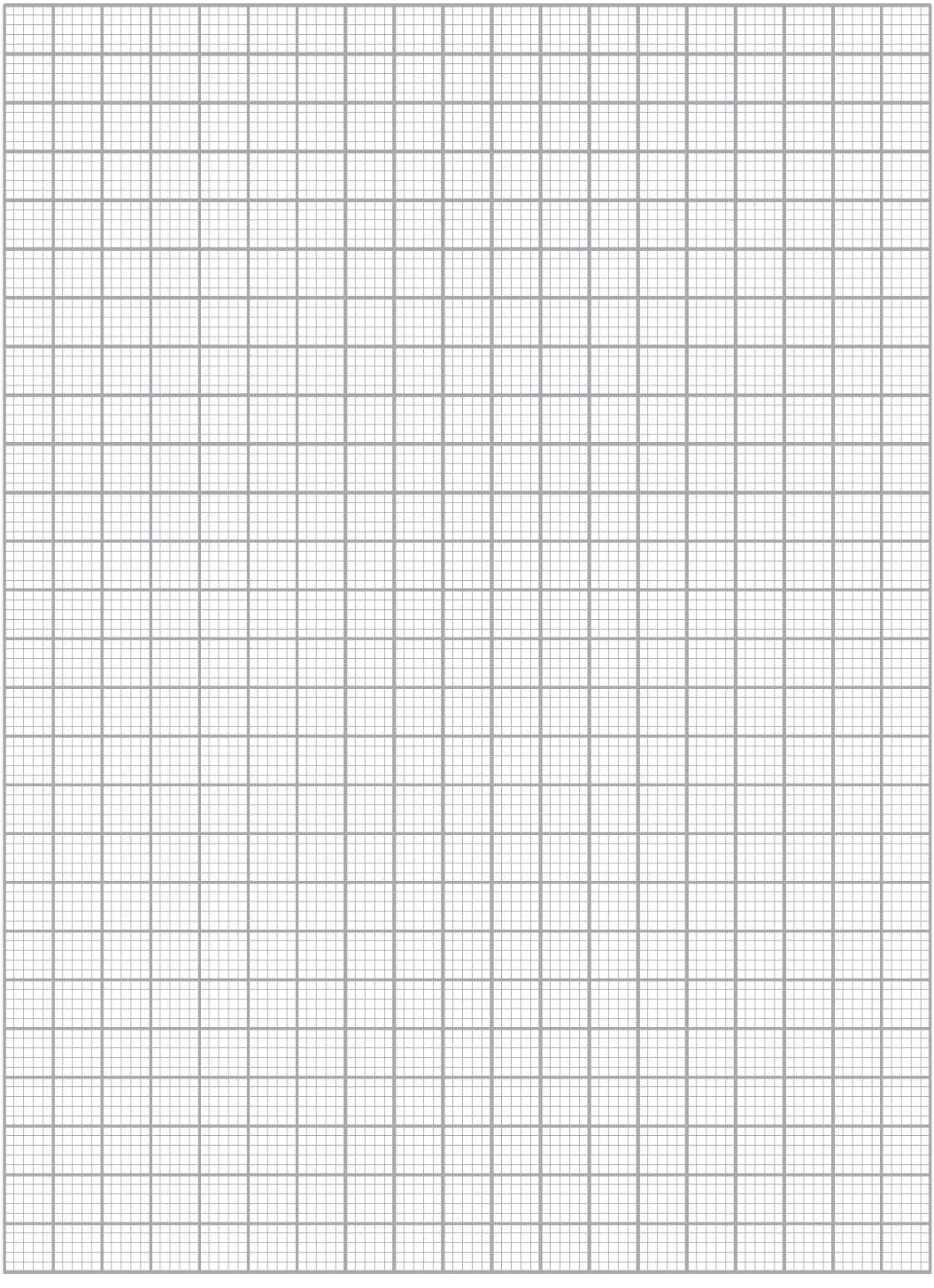
Distance (d) of hairdryer from shoot/cm	Wind strength (1/d)/arbitrary units	Rate of bubble movement / mm min^{-1}	
		Cool setting on hairdryer	Hot setting on hairdryer
11	0.09	5.2	3.8
14	0.07	4.1	4.9
20	0.05	3.1	6.2
25	0.04	2.4	5.1
50	0.02	1.0	2.3

(a) Using the most appropriate graphical technique, plot the above data for the title:

'*How the rate of bubble movement in a potometer containing a sycamore shoot is affected by wind strength at two different environmental temperatures*.'

Note: You do not need to include the title on the graph. (4 marks)

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(b) Describe and explain the trend shown by the results for the hairdryer on the cool setting. (3 marks)

(c) Describe ***two*** ways in which the trends shown for the two heat settings differ and give an explanation for each difference. (4 marks)

##

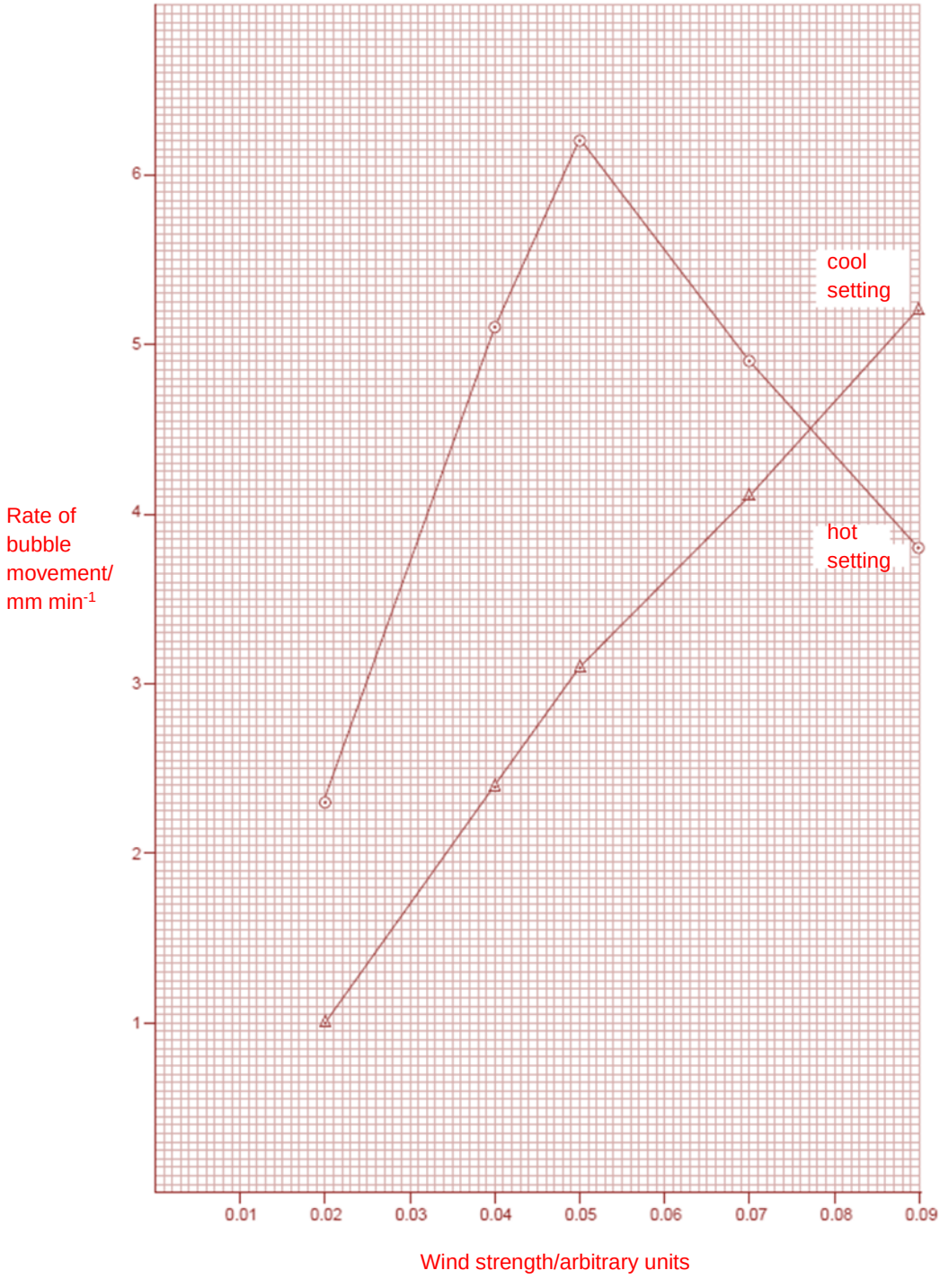
(a) Axes the correct way round (wind strength on x-axis) and labelled including units (1)

Selection of appropriate scale (1)

Points plotted accurately and joined by short straight lines (1)

Lines labelled/legend included (1)

How the rate of bubble movement in a potometer containing a sycamore shoot is affected by wind strength at two different environmental temperatures



(b) As the wind strength increases the rate of bubble movement increases. (1)

Any ***two*** of the following: (2)

- Water vapour blown away from leaf surface (1)
- Water vapour diffusion gradient increases (1)
- Pull of water into the shoot increases (1)

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(c) **Difference:** At hot setting, the rise in rate of bubble movement at lower wind strengths is steeper than at cool setting. (1)
Explanation: Higher temperatures cause increased rate of water evaporation. (1)
Difference: At hot setting, the rate of bubble movement decreases after a certain wind strength but with cool setting it keeps on rising. (1)
Explanation: Hot setting triggers closure of stomata to prevent desiccation. (1)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Presenting information	4
(b)	Analysing data	1
	Applying concepts	2
(c)	Analysing data	2
	Applying concepts	2

|!|BCELC0110000022|!| **SI**

CCEA GCE AS Biology Assessment Unit AS 2 Jun 2015 Q7

The stomatal density (number of stomata per unit area) found on leaves may be influenced by the plant's habitat.

An investigation was carried out to determine the stomatal density in four different plant species. The number of stomata on 4.5 mm^2 of leaf lower epidermis was counted.

The table below shows the results.

Species	Number of stomata	Stomatal density / stomata mm^{-2}
A	26	5.78
B	40	8.89
C	37	
D	20	4.44

(a) (i) Calculate the stomatal density for species C. (1 mark)

(ii) Identify which of the species investigated is most likely to grow in a dry environment. Explain your answer. (2 marks)

(b) (i) Define the term transpiration. (2 marks)

The leaves of plants growing in dry environments often have microscopic leaf hairs extending from their lower epidermis.

(ii) Explain the advantage of this. (3 marks)

##

(a) (i) 8.22 stomata mm^{-2} (1)

(ii) D (1)

Fewer stomata reduces the loss of water by transpiration. (1)

(b) (i) The evaporation of water from the mesophyll surface / leaf cells (1)

the diffusion of the water (vapour) into the atmosphere via the stomata / cuticle (1)

(ii) Hairs disrupt air flow over leaf surface / traps air (1)

Creation of humid atmosphere (1)

reduces diffusion / transpiration / evaporation. (1)

(8 marks)

##

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	Skill assessed	Mark(s)
(a)(i)	Mathematical skills	1
(a)(ii)	Analysing data	1
	Applying concepts	1
(b)(i)	Understanding basic concepts	2
(b)(ii)	Applying concepts	3

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